Chapter 3.

§3.1.

$\frac{d}{dx}[c] = 0$

$\frac{d}{dx}[x^r] = r x^{r-1}$ → General Power rule.

$\frac{d}{dx}[c f(x)] = c \frac{d}{dx}[f(x)]$.

$\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}[f(x)] \pm \frac{d}{dx}[g(x)]$

Let $f(x) = a^x$, then $f'(x) = f'(0) a^x$

The number e satisfies $\lim_{h \to 0} \frac{e^h - 1}{h} = 1$ $\Big\} \Rightarrow \frac{d}{dx}[e^x] = e^x$

§3.2.

Product Rule: $\frac{d}{dx}[f(x)g(x)] = f(x)\frac{d}{dx}[g(x)] + g(x)\frac{d}{dx}[f(x)]$

or $(f(x)g(x))' = f(x)g'(x) + g(x)f'(x)$.

Quotient Rule: $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)\frac{d}{dx}[f(x)] - f(x)\frac{d}{dx}[g(x)]}{[g(x)]^2}$

or $\left(\frac{f(x)}{g(x)}\right)' = \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}$.

§3.3.

$$\boxed{\lim_{x \to 0} \frac{\sin x}{x} = 1}$$

$\frac{d}{dx}(\sin x) = \cos x$, $\frac{d}{dx}(\cos x) = -\sin x$.

$\frac{d}{dx}(\tan x) = \sec^2 x$, $\frac{d}{dx}(\cot x) = -\csc^2 x$

$\frac{d}{dx}(\sec x) = \sec x \tan x$, $\frac{d}{dx}(\csc x) = -\csc x \cot x$

§3.4.

Chain Rule: Let $F(x) = f(g(x))$, then

$$F'(x) = f'(g(x))\, g'(x).$$

or

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} \quad \text{where } y = f(u),\ u = g(x).$$

Power Rule + Chain Rule

$$\frac{d}{dx}[u^n] = n u^{n-1}\frac{du}{dx}.$$

or

$$\frac{d}{dx}[(g(x))^n] = n\,(g(x))^n g'(x)$$

Exponential function: $\frac{d}{dx}[a^x] = a^x \ln a.$

Exponential Rule + Chain Rule

$$\frac{d}{dx}[a^u] = a^u \ln a \,\frac{du}{dx}$$

or

$$\frac{d}{dx}[a^{g(x)}] = a^{g(x)}(\ln a)\, g'(x)$$

§3.5.

Implicit Differentiation: $f(x,y) = g(x,y)$

$$\Rightarrow \frac{d}{dx}[f(x,y)] = \frac{d}{dx}[g(x,y)]$$

$$\frac{\partial f}{\partial x} - \frac{\partial f}{\partial y}\boxed{\frac{dy}{dx}} = \frac{\partial g}{\partial x} - \frac{\partial g}{\partial y}\boxed{\frac{dy}{dx}}$$

where $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ are like regular derivatives.

Method using Implicit Differentiation to find the direvative $\frac{dy}{dx}$ for a given curve defined implicitly.

Derivatives of Inverse Trigonometric functions.

$$\frac{d}{dx}[\sin^{-1}x] = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}[\tan^{-1}x] = \frac{1}{1+x^2}, \quad \frac{d}{dx}[\sec^{-1}x] = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}[\cos^{-1}x] = -\frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}[\cot^{-1}x] = \frac{-1}{1+x^2}, \quad \frac{d}{dx}[\csc^{-1}x] = \frac{-1}{x\sqrt{x^2-1}}$$

§3.6.

logrithmic function: $\underline{\frac{d}{dx}[\log_a x] = \frac{1}{x\ln a}}$. $\rightarrow \frac{d}{dx}[\ln x] = \frac{1}{x}$.

<u>logarithmic Rule + Chain Rule</u>

$$\underline{\frac{d}{dx}[\ln u] = \frac{1}{u}\frac{du}{dx}}$$

$$\underline{\frac{d}{dx}[\ln|x|] = \frac{1}{x}.}$$

or

$$\frac{d}{dx}[\ln g(x)] = \frac{1}{g(x)} g'(x)$$

Some better rules:

$$\frac{d}{dx}\left[\ln\frac{f(x)}{g(x)}\right] = \frac{d}{dx}[\ln f(x) - \ln g(x)]$$

$$\frac{d}{dx}[\ln(f(x)g(x))] = \frac{d}{dx}[\ln f(x) + \ln g(x)]$$

<u>logarithmic Differentiation</u>

$$y = f(x).$$

1. Take natural logarithmic differentiation on both sides.

$$\ln y = \ln f(x)$$

2. Differentiate implicitly with respect to x.

$$\frac{d}{dx}[\ln y] = \frac{d}{dx}[\ln f(x)] \rightarrow \left(\frac{1}{y}\right)y' = \frac{d}{dx}[\ln f(x)]$$

3. Solve the resulting equation for y'.

The number e as limit: $\underline{e = \lim_{x\to 0}(1+x)^{\frac{1}{x}}}$.

§3.9.

See Problems in Page 241-245.

Chapter 4.

§4.1.

Definitions for absolute maximum, absolute minimum (extreme values)
local maximum, local minimum
for a given function $y = f(x)$.

Fermat Theorem: f has a local maximum/minimum at C exists
$\Rightarrow f'(c) = 0$.

$\boxed{\text{critical number}}$: $f'(c) = 0$ or $f'(c)$ doesn't exist.

The closed interval Method (The extreme values of f on $[a,b]$).

1. Find the values of f at all critical numbers of f in (a,b).
2. Find the values of f at a and b.
3. The largest one from steps 1 & 2 → the absolute maximum value.
   The smallest one → the absolute minimum value

§4.2.

The Rolle's Theorem

i) f is continuous on $[a,b]$
ii) f is diffentiable in (a,b)
iii) $f(a) = f(b)$

$\Rightarrow$ There exists a number c in (a,b) such that

$$f'(c) = 0$$

The Mean-Value Theorem

i) f is continuous on $[a,b]$
ii) f is differentiable in (a,b)
$\Rightarrow$ There exists a number c in (a,b) such that

$$f'(c) = \frac{f(b)-f(a)}{b-a}.$$

i.e.

$$f(b)-f(a) = f'(c)(b-a)$$

§4.3.

Increasing/decreasing Test. P287

The first derivative Test P288.

Definitions for **concavity**: $f''(x)>0 \rightarrow$ concave upward
$f''(x)<0 \rightarrow$ concave downward.

Inflection points

The second derivative Test P292.

1) $f'(c)=0$, $f''(c)>0 \Rightarrow$ a local minimum at c
2) $f'(c)=0$, $f''(c)<0 \Rightarrow$ a local maximum at c.